Reviews of 8159 - *"Human–LLM Collaborative Pipeline for Efficient Audio Effect Parameter Tuning"*

Reviewer 4 (1AC)

Expertise

Knowledgeable

Originality (Round 1)

Medium originality

Significance (Round 1)

Low significance

Research Quality (Round 1)

Low research quality

Recommendation (Round 1)

I recommend Reject

1AC: The Meta-Review

Overall Recommendation: While the three reviewers (2AC, R1, R2) saw promise in the idea of using an LLM to translate natural language descriptions into parameter tuning for a digital audio workstation (DAW), especially for novice users, all three reviewers felt that the evaluation fell far short of a CHI paper. Thus, I am recommending rejection, but encourage the authors to consider whether a reconfigured evaluation study and much more detailed description of the methods and results of such a study might enable the paper to better reflect the research's contributions in the future.

Key Strengths: All three reviewers felt that the idea of using natural language descriptions to tune audio effect parameters in the presented HCAP system would be compelling. 2AC and R1 articulate that this system would be especially useful for novice users, who might lack the technical insight about how to transforms their goals into effect parameter settings.

Key Weaknesses: Unfortunately, all three reviewers felt that the evaluation section of the paper, both the presentation of the methods and the presentation of the results, fell well below the standards for CHI publications. For instance, the reviewers had important questions that the paper did not answer related to the precise task activities during the user study, the goal given to participants for the user study, participant recruitment, the specific data that was collected, how the data was analyzed, the basics of whether responses were collected from surveys or an interview, and more. For instance, R2 was left confused by the inclusion of participant quotes when the paper seemed primarily to reference a survey as the data source.

The reviewers suspect that the shortcoming might not just be an issue in reporting. R1, for instance, highlights how an ablation study would be needed to pinpoint the particular contributions of different design decisions. All reviewers felt that the inclusion of more objective metrics would be important.

All three reviewers also give specific suggestions about important prior work that is missing, and it will be necessary to place this paper into the existing literature including those references.

I also want to highlight 2AC's points about clarifying the novelty of the contributions of HCAP. For instance, the ultimate UI seems very similar to traditional DAW interfaces. While the use of natural language descriptions to modify audio effect parameters is novel, what is the contribution of this work (as opposed to the contributions of the designers of LLMs)? As 2AC questions, what are the "unique considerations...required to design such a system for audio parameter control"?

Reviewer 3 (2AC)

Expertise

Expert

Originality (Round 1)

Low originality

Significance (Round 1)

Low significance

Research Quality (Round 1)

Low research quality

Contribution Compared to Length (Round 1)

The paper was too short to address its claimed contribution.

Figure Descriptions

The figure descriptions are adequate and follow the accessibility guidelines.

Recommendation (Round 1)

I recommend Reject

Review (Round 1)

This paper introduces HCAP, a DAW plug-in for audio effect parameter tuning. The system uses an LLM to generate parameter recommendations based on users' natural language descriptions, audio examples, and preference history. A technical evaluation and a user study were briefly presented.

Overall, I appreciate the direction of using AI to help users achieve their desired sound in DAWs. This has been a long-standing challenge, especially for novices who struggle to map high-level musical concepts to low-level parameters. In that regard, I am glad to see more work exploring how emerging techniques can be applied. That said, I do not consider the current paper suitable for publication at CHI, nor do I think a revise-and-resubmit would bring it to an acceptable level. I therefore recommend rejection, although this might make a good demo at CHI. Below is my reasoning.

First, I see little novelty in either the technical pipeline or the UI. The pipeline is a straightforward wrapper around an LLM, and the paper offers limited insight into what unique considerations are required to design such a system for audio parameter control. The design feels shallow—something that could have been achieved in a short hackathon project. I am also surprised that the system does not employ multimodal LLMs, given that representing audio information in text alone may lose crucial details. The UI likewise lacks novelty: it looks almost identical to traditional DAW interfaces. Given that the paper opens by critiquing existing DAW paradigms, it is inconsistent that the proposed interface follows the same paradigm.

The evaluation methodology and reporting do not meet CHI's standards of rigor. Both sections are extremely brief, leaving many questions unanswered. For example, in the technical evaluation, the two baselines—manual adjustments by professional users and expert-curated presets—are not clearly differentiated. What's the difference between pro users and experts in this context? The user study description is similarly short and lacks methodological clarity.

Assisting users with parameter tuning has been an active research topic in HCI, yet the paper does not adequately engage with prior work. Important literature in audio parameter control (e.g., [1, 2]) is missing. The related work section is unfocused; for instance, the “AI-Assisted Creative Tools” section is overly broad and cites only a handful of examples out of a large possible body of work. The authors should instead focus on more directly relevant areas, such as interactive parameter tuning in creative applications, including those beyond audio (e.g., visual generation tools).

In summary, this paper falls significantly below the bar for a CHI full paper. I do not believe a revision would address the fundamental issues, so I recommend rejection.

[1] Active learning of intuitive control knobs for synthesizers using gaussian processes (IUI ‘14)

[2] SynthScribe: Deep Multimodal Tools for Synthesizer Sound Retrieval and Exploration (IUI ‘24)

Reviewer 1 (reviewer)

Expertise

Expert

Originality (Round 1)

Low originality

Significance (Round 1)

Low significance

Research Quality (Round 1)

Low research quality

Contribution Compared to Length (Round 1)

The paper was too short to address its claimed contribution.

Figure Descriptions

The figure descriptions are adequate and follow the accessibility guidelines.

Recommendation (Round 1)

I recommend Reject

Review (Round 1)

#8159: Human–LLM Collaborative Pipeline for Efficient Audio Effect Parameter Tuning

The contribution of this paper is an artifact for assisting Digital Audio Workstation (DAW) users to automatically translate instructions in natural language to manipulate DAW parameter-level controls. For this, they develop a human-LLM collaborative pipeline called HCAP, which consists of multiple stages. The authors evaluate their artifact and pipeline using objective metrics and a user study.

While the authors research an important topic (translating language instructions to DAW controls), as a reader, I found the text difficult to read. Some of the motivations, such as the shift of cognitive resources from creative exploration to manual trial and error manipulation while using DAW, are clear. However, there are too many jargons and other unexplained and unreferenced theories (for instance, what does germane load or schema refinement mean? Similarly, what is the propose-and-dispose model?) introduced in the text that need clarity and explanation. Furthermore, the experiments evaluating objective metrics and the user study are insufficient and lack rigor. Hence, my recommendation for rejection for this cycle of the conference. Please see details below.

Scope:

On line 148, the authors note "Yet little work addresses load management under real-time latency[…] across diverse expert levels […] balancing automation support with skill development". The authors are trying to take on too many things to study/address. Narrowing down the scope will be beneficial.

Furthermore, the paper's scope in terms of its target users is not well defined. Is the system targeted for professional sound designers or novice users? The authors correctly point out that the needs of both these groups are different. Identifying the target users of the system will be helpful in better contextualizing the design decisions and user studies.

Text that needs more clarity:

1. Abstract Line 10: What is preference-aware fusion and modality-aware fusion? The abstract outlines to solve the problem of translating language instructions to DAW controls - articulating how did this goal evolves into these requirements will assist the

reader to understand the paper's motivations and design better.

2. Line 30, 39, etc: As mentioned before, elaborating more on Cognitive Load Theory (CLT) and explicating what germane load and schema refinement mean would be helpful. Also, self-determination theory (Line 157). Also, add relevant references.

3. Partially true claim: (Line 132: "Parameter-level creative collaboration therefore remains underexplored") This is partially true. Before text-based controls, all generative audio AI interactive interfaces were controlled by parameters only (e.g., using pitch, tempo, etc). Perhaps the authors mean interfaces designed post-LLM integration?

4. Line 137: What are Miller's findings? Outline in a sentence or two. Providing a snapshot would assist readers unfamiliar with the topic.

5. Line 160: What are categorical features and dynamic context prompting?

6. Line 162: What is the propose-and-dispose interaction model?

System Design:

1. Line 188: What is the structure of the text descriptors? Adding the template to the appendix would be helpful.
2. Line 189: How are the parameters technically inferred from the exemplar?
3. Line 199: Specialist prompting and candidate parameter sets need more details.

Evaluation:

1. The biggest concern I have is that the user study is insufficient and lacks rigor. How were the participants recruited? What were the demographic details? How were they compensated? What was the creative goal given to the participants while performing the task for the study? What kind of data was collected? Was there an observational analysis or an analysis of interview transcripts only? How were the themes developed? Why wasn’t NASA-TLX (elaborated previously) used in this study? Just to name a few.
2. Also, the objective experimental setup needs more details - for instance, details on how the guitar track was recorded, the complexity of the track, the experience level of the artist recording it, etc, are missing.

3. Also, why were LSD and harmonic similarity chosen as objective metrics?
4. Lack of ablation studies - for instance, how does the system perform if no examples are given? Similarly, if no text and only examples are given?

References:
The authors might find grounding their work in mixed-initiative creative interfaces and generative AI interaction design principles useful for this work.

Sebastian Deterding et al. Mixed-initiative creative interfaces (2017)

Justin D. Weisz et al. Toward General Design Principles for Generative AI Applications (2023)

To summarize, while the authors suggest an interesting area of research, I believe the aforementioned issues, especially surrounding clarity of the manuscript and rigor in the user studies and evaluation, need to be addressed. As such, I do not think this can be accomplished in the short RR period. I hence recommend rejection for this cycle of the conference.

Reviewer 2 (reviewer)

Expertise

Expert

Originality (Round 1)

Medium originality

Significance (Round 1)

Low significance

Research Quality (Round 1)

Low research quality

Contribution Compared to Length (Round 1)

The paper length was commensurate with its contribution.

Figure Descriptions

The figure descriptions are adequate and follow the accessibility guidelines.

Recommendation (Round 1)

I can go with either Reject or Revise and Resubmit

Review (Round 1)

Thank you for allowing me the opportunity to review this work. Below, I have included my full review of the piece and I hope the comments can help strengthen the argument in the paper.

Throughout my review I have used the notation (L#) to indicate the line number that I am referring to.

Summary of the Article
This paper introduces the Human-LLM Collaborative Pipeline (HCAP), a system designed to simplify the complex process of tuning audio effect parameters in Digital Audio Workstations (DAWs). The main contributions of this work include a three-stage system architecture that captures user intent (through text or audio references), fuses it with user preferences to generate context-aware parameter suggestions, a dual-interface interaction design that combines a conversational panel for natural language input with traditional DAW controls, and an empirical evaluation and user study for the HCAP system.

Positive Aspects
The paper reads well overall and the use of figures as an explanatory medium are extremely helpful to the reader! The gap is extremely well defined and well motivated for the work that is being conducted.

Major Problems
The most glaring problem that I see is that of a focus problem on the system design as opposed to the user evaluation. Although the system architecture was well documented, the paper lacked a rigorous evaluation regarding user evaluation and it unfortunately hurt the arguments made throughout the piece. I've tried to cluster my critiques a bit here to hopefully help the authors strengthen the arguments made throughout this piece.

System Design:
The system design is made clear throughout the work, but there are a number of inputs/outputs from stage to stage that can become confusing to the reader after the second stage. I would encourage the authors to explore the use of a Logic Model which helps to diagram these inputs/activities/outputs for each stage in a figure which might be illustrative for the reader as an overview of the full pipeline before going into more detail for each stage. Figure 1 does a great job with the full process, but I think

so much gets lost in what the three stages are doing that it undermines the figure a bit in the sake of clarity.

Reference: Taylor-Powell, Ellen, and Ellen Henert. "Developing a logic model: Teaching and training guide." _Benefits_ 3.22 (2008): 1-118.

However, I am worried generally about the work overall since I see a number of references missing that are very much similar to this line of work such as the following papers:
- Chu, Annie, et al. "Text2fx: Harnessing clap embeddings for text-guided audio effects." _ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)_. IEEE, 2025.
- Doh, Seungheon, et al. "Can Large Language Models Predict Audio Effects Parameters from Natural Language?." _arXiv preprint arXiv:2505.20770_ (2025).
- Stables, Ryan, et al. "SAFE: A system for the extraction and retrieval of semantic audio descriptors." (2014): 1-2.

There are others in the space too and it would be helpful to the reader if the authors motivate their work alongside these other works that have attempted similar feats. These papers are not specifically address this problem in the case of DAWs, and I believe the inclusion of the other works strengthens this work.

There is not much detail for the database of audio references or user preferences, but this seems like case-based reasoning would be maybe an appropriate use for this? There are many details omitted throughout this piece of the system. I believe putting some of the details in the appendix and leaving the main body of the paper to explore the user evaluation would be helpful to strengthen the claims in this work.

User Evaluation:
This section, in my opinion, is the reason I would recommend a revise and resubmit or reject for this paper. There's a missed opportunity here in talking about how/why the system was created and what method was used in the creation of this: convergent design, explanatory sequential design, exploratory sequential design, etc. This would help better inform the data analysis methods that were chosen and the rigor to which each analysis should be conducted.

Reference: Creswell, John W., and Vicki L. Plano Clark. _Designing and conducting mixed methods research_. Sage publications, 2017.

There is no mention of the experimental setup, how participants were compensated (if they were), how participants were recruited, nor how long the experiment lasted (since the system was learning over time what the user preferred). Although the demographics containing the expertise of the participants were included, the omission of the other details makes it difficult to understand many nuances of the experiment, which leads to the reader questioning conclusions such as "workflow efficiency" since this would be more indicative of task efficiency, as most users would using a traditional DAW for this workflow instead of a separate tool.

There are a number of places where there are quotes used in the piece such as "eye-opening" (L626) or "felt more personal over time" (L631) but from what I read in the Appendix, I could not find a place where the users were able to write in responses since most of this was in Likert form or scales and either multiple or single choice questions. Were interviews used? If so were they unstructured, semi-structured, or structured? Broadly, I found the Findings section to be representative of a thematic analysis section with all the quotes throughout the paragraphs, but it's still vague where these responses are coming from.

Data Wrangling:
Throughout these sections too, I found it a bit jarring to read things such as "71.43% wanting inspectable preference archives..." or "92.85% specifically mentioned that tone adaptation...". In this section, the authors might want to use the number of participants instead of percentages such as "Nearly all participants (N=13) mentioned that tone adaptation..." The authors identified N=14 as the number of participants to this user evaluation, and the percentages distract the reader from reading the arguments made. The reason for this as well is in the use of the mean for Likert data which is not always useful depending on the type of Likert data collected. For example with "(mean rating 4.0/5)" (L615), from the data gathered, I see the median being more representative.

Reference: Boone Jr, Harry N., and Deborah A. Boone. "Analyzing likert data." _The Journal of extension_ 50.2 (2012): 48.

Future Work:
The future work talked mainly about what features would be added, but I felt there was a lack of connection with the rest of the paper and the findings in this work. There were key findings the authors highlighted in the work, but they were omitted in this section in favor of explaining more of the features that would be implemented. I think adding and expanding on the findings from the user evaluations in terms of research would be of great benefit in strengthening the arguments made throughout this work.

Minor Comments
- (L29) spell out MIDI, (L677-687) VST, AU. Many folks will not know these acronyms for this venue.
- (L286) add at least one cite for the claim "a gap well documented in HCI research..."
- (L349, L350) sentence is doubled. Just delete one of them

Return to submission and reviews